

# E-Commerce Analytics

Business Insights Report — Phase 3 (EDA, ML & Statistical Analysis)

This report summarizes findings from six analyses run on the company's e-commerce dataset (Jan 2023 – Dec 2025): exploratory analysis, churn prediction, revenue forecasting, anomaly detection, an A/B test read-out, and RFM customer segmentation. Each section states the business question, the finding, and the recommended action.

## Headline Numbers

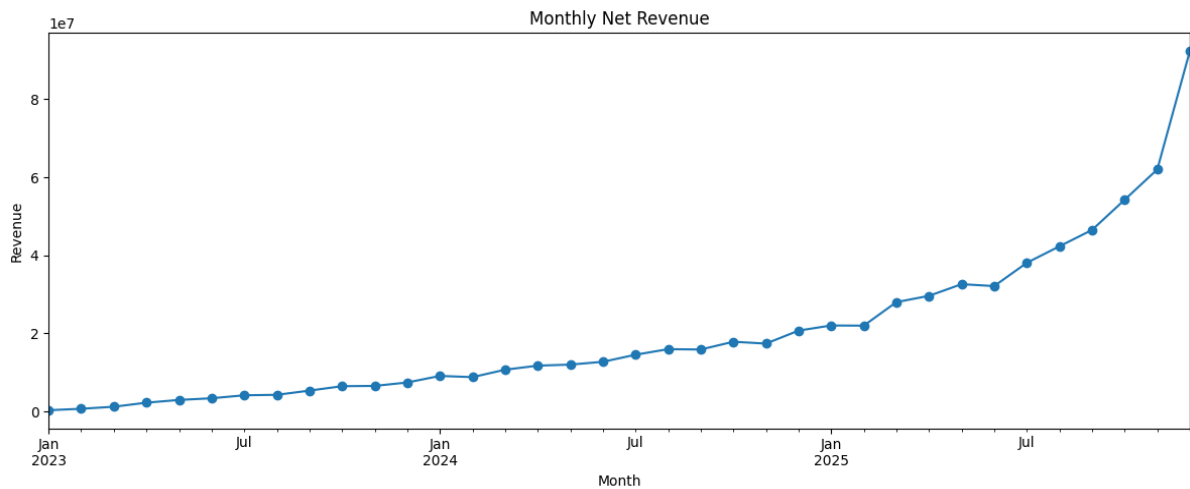
|                                    |                                   |                                |
|------------------------------------|-----------------------------------|--------------------------------|
| <b>15,000</b><br>Total Customers   | <b>34,254</b><br>Completed Orders | <b>\$713.9M</b><br>Net Revenue |
| <b>\$20,841</b><br>Avg Order Value | <b>\$39.4M</b><br>Marketing Spend | <b>\$55.8M</b><br>Refunds Paid |

## Executive Summary

- Revenue is growing but the growth curve is accelerating, not linear — monthly net revenue climbed from under \$1M (early 2023) to over \$90M (Dec 2025). Growth-stage planning (inventory, staffing, cash flow) should assume continued acceleration, not a flat trend line.
- 51% of customers who have ever purchased are currently churned (no completed order in 90+ days). The churn model (Random Forest, ROC-AUC 0.70) ranks 407 high-spend customers as high-risk — this is the priority retention list.
- The checkout redesign A/B test is a clear, statistically significant win: +2.28 percentage points conversion uplift (95% CI entirely positive). Recommendation: SHIP.
- RFM segmentation shows the top two segments (Champions + Loyal Customers, 4,521 customers) generate the large majority of revenue — retention spend should be weighted toward protecting these segments, not just acquiring new ones.
- A 90-day linear revenue forecast is included, but given the accelerating trend noted above, treat it as a conservative floor, not a target — see forecasting section for the important caveat.

# 1. Business Overview — Exploratory Analysis

Question: What does the underlying business look like — revenue trend, order health, acquisition mix, and where the money comes from?

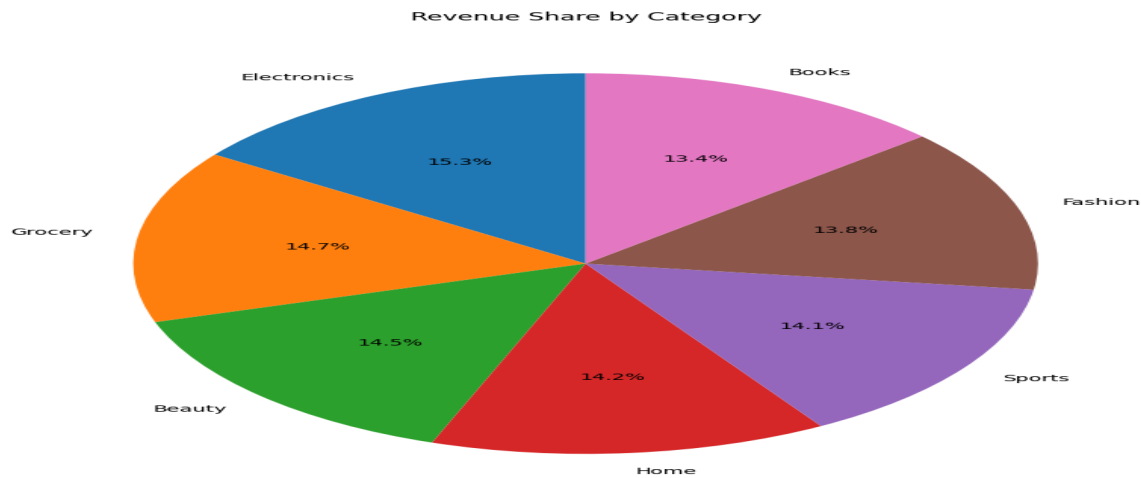


Monthly net revenue, Jan 2023 – Dec 2025. Growth is accelerating, not linear — the last 6 months alone account for a large share of total 2025 revenue.

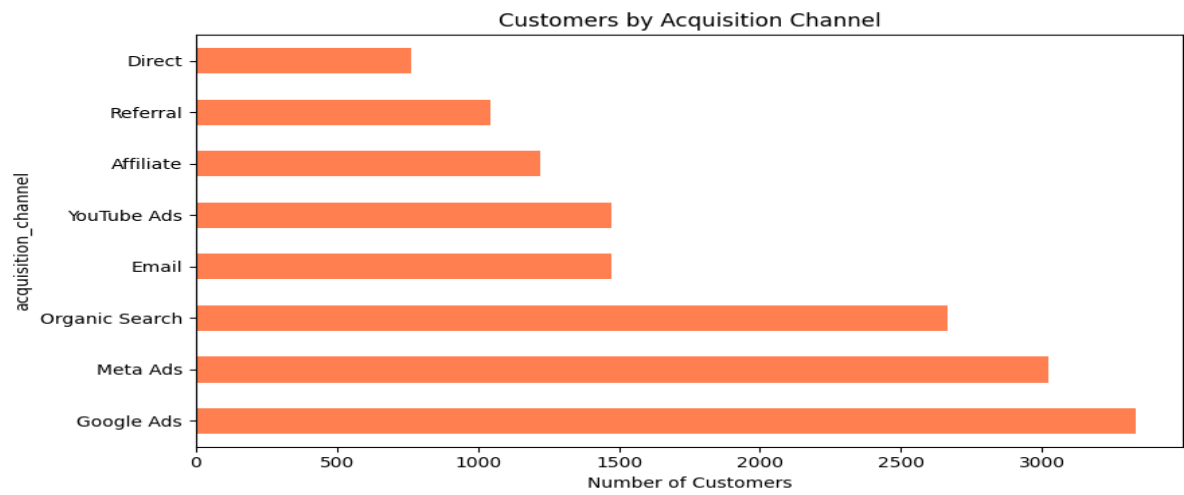
## Order status breakdown:

| Status    | Orders | % of total |
|-----------|--------|------------|
| Completed | 34,254 | 57.1%      |
| Returned  | 8,692  | 14.5%      |
| Pending   | 8,541  | 14.2%      |
| Cancelled | 8,498  | 14.2%      |

Cancelled and Pending orders together account for 28.4% of all orders placed — worth a dedicated checkout/fulfillment funnel review, since every Cancelled/Pending order is potential revenue never captured.



Revenue share by product category — fairly even split (no single category dominates), reducing concentration risk.



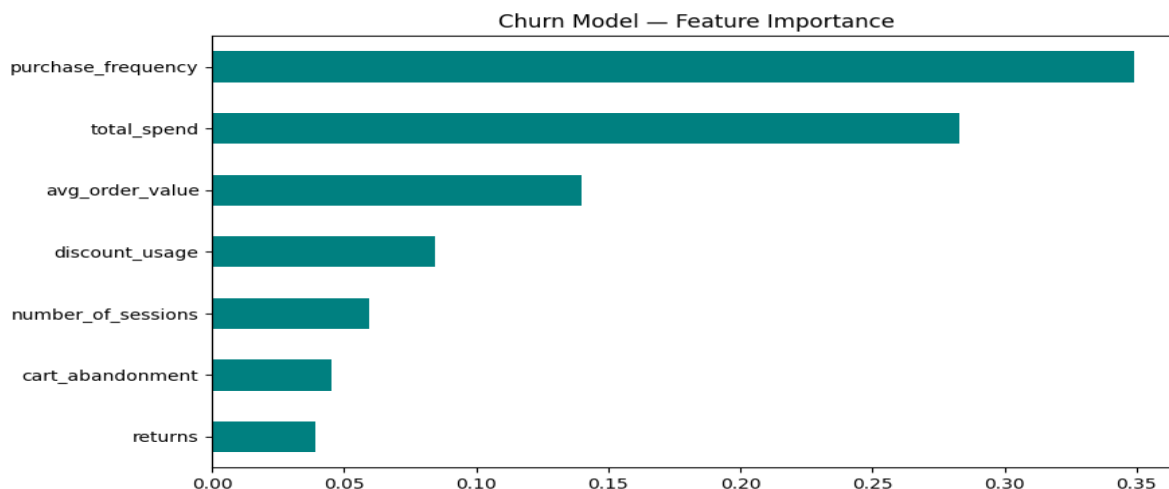
Customers by acquisition channel.

## 2. Customer Churn Prediction

Question: Which customers are at risk of leaving, and who should retention spend target first? Churn is defined as no completed order in the last 90 days, among customers who have purchased before.

|                                                |                                   |                              |
|------------------------------------------------|-----------------------------------|------------------------------|
| <b>10,436</b><br>Customers w/ purchase history | <b>51.4%</b><br>Currently churned | <b>0.70</b><br>Model ROC-AUC |
|------------------------------------------------|-----------------------------------|------------------------------|

A Random Forest classifier was trained on behavioral features only — purchase frequency, total spend, average order value, site sessions, cart abandonment, returns, and discount usage. Recency itself was deliberately excluded as a feature since it is how churn is defined (using it would leak the answer into the model).



What predicts churn: purchase frequency and total spend are by far the strongest signals — low-frequency, low-spend customers churn most.

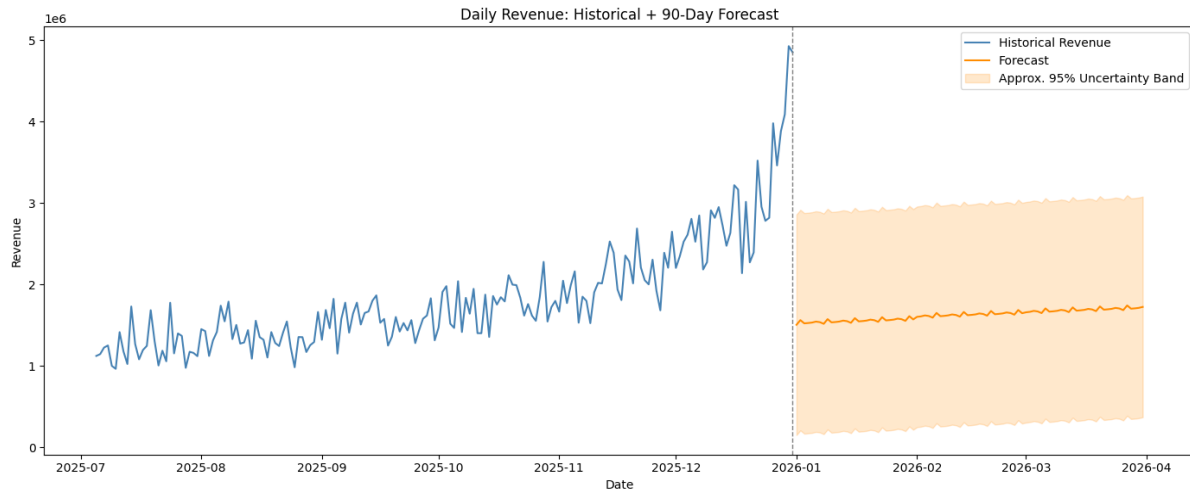
Business action: 407 customers are flagged as both high-value (above-median total spend) and high churn-risk. This is the priority outreach list (see `retention_priority_list.csv`) — targeted win-back offers here protect the most revenue per retention dollar spent.

Caveat: precision/recall on the minority class is moderate (~0.63-0.70), typical for behavioral churn models. Use the risk score as a prioritization signal to guide outreach, not as a certainty about any individual customer.

### 3. Revenue Forecasting (Next 90 Days)

Question: Based on trend and seasonality, what revenue should we plan for over the next 30/60/90 days?

|                                           |                                           |                                            |
|-------------------------------------------|-------------------------------------------|--------------------------------------------|
| <b>\$46.6M</b><br>Next 30 days (forecast) | <b>\$95.6M</b><br>Next 60 days (forecast) | <b>\$146.4M</b><br>Next 90 days (forecast) |
|-------------------------------------------|-------------------------------------------|--------------------------------------------|



Historical daily revenue (blue) vs. 90-day linear-trend forecast (orange), with an approximate uncertainty band.

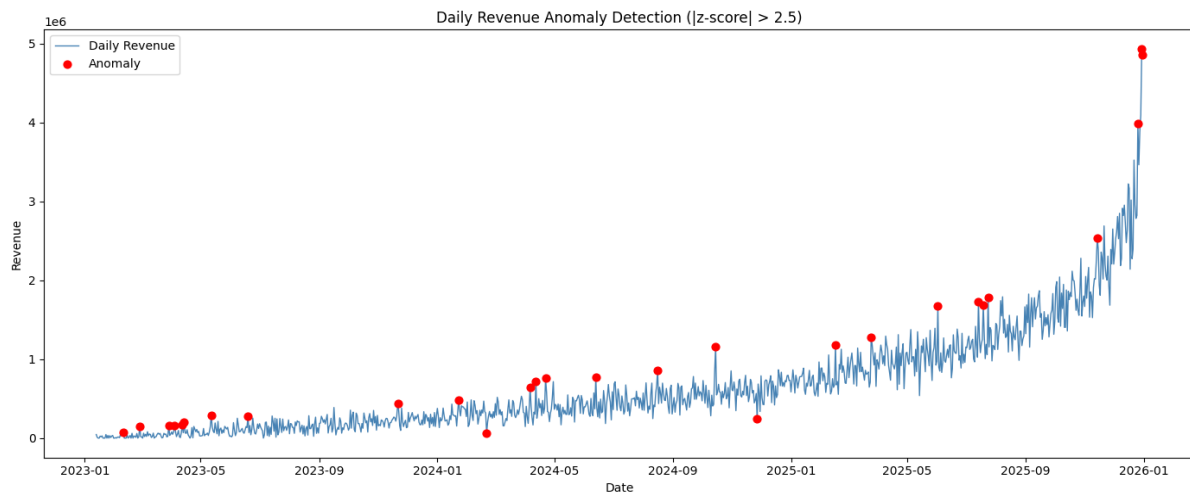
**Important caveat:** this model uses a linear trend + day-of-week/month seasonality. The actual revenue curve in Section 1 is accelerating (compounding growth), which a straight line under-predicts — visible in the chart above, where the forecast sits noticeably below the most recent daily actuals. Backtest error on the last 60 days was high (MAPE ~49%). Treat this forecast as a conservative planning floor, not a target, and prioritize a growth-curve or ML-based model (e.g. Prophet, gradient boosting) for finance-grade forecasting.

## 4. Anomaly Detection

Question: Which days had unusual revenue, order volume, marketing spend, or refunds that deserve investigation?

Method: rolling 30-day z-score, flagged when  $|z| > 2.5$ .

|                                |                                     |                               |
|--------------------------------|-------------------------------------|-------------------------------|
| <b>28</b><br>Revenue anomalies | <b>24</b><br>Order volume anomalies | <b>28</b><br>Refund anomalies |
|--------------------------------|-------------------------------------|-------------------------------|



Daily revenue with anomalous days flagged in red.

Largest flagged revenue drop: 2024-11-27 (z-score -2.59). Order volume that day was not equally anomalous, suggesting the drop is driven by average order value / discounting / product mix rather than site traffic — worth cross-checking against the marketing calendar for that date.

Business action: route the flagged dates in anomalies\_\*.csv to the relevant team (ops for order-volume drops, finance for spend spikes, product/QA for refund spikes) as a starting investigation list, not a root-cause conclusion.

## 5. A/B Test — Checkout Redesign

Question: Does the redesigned checkout flow improve conversion enough to justify shipping it?

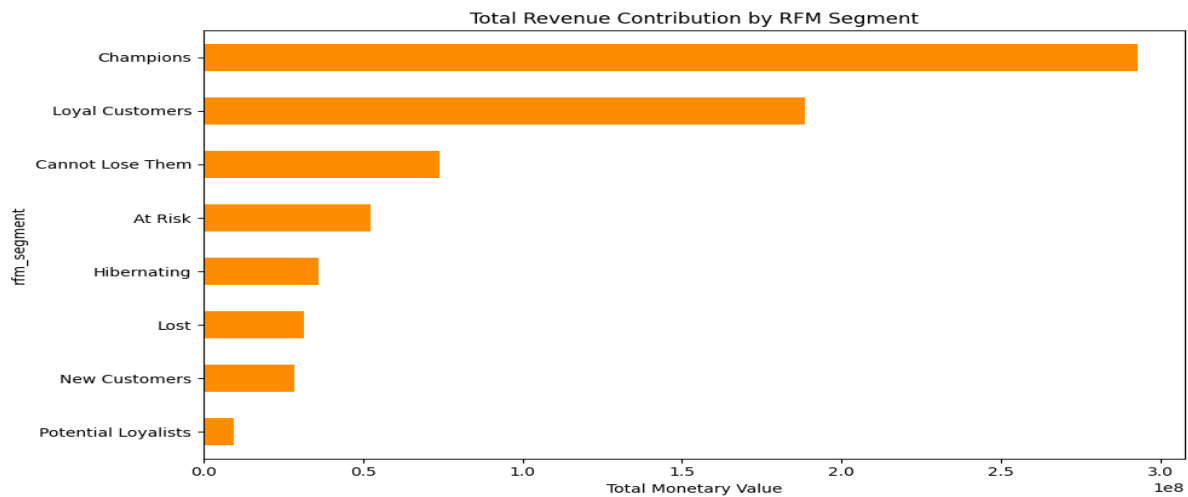
|                                     |                                       |                           |
|-------------------------------------|---------------------------------------|---------------------------|
| <b>11.73%</b><br>Control conversion | <b>14.01%</b><br>Treatment conversion | <b>+2.28 pp</b><br>Uplift |
|-------------------------------------|---------------------------------------|---------------------------|

| Metric                                     | Value           |
|--------------------------------------------|-----------------|
| Sample size (control / treatment)          | 3,981 / 4,019   |
| P-value                                    | 0.0024          |
| 95% Confidence interval on uplift          | [0.81, 3.74] pp |
| Statistically significant?                 | Yes             |
| Business significant? ( $\geq 1.0$ pp bar) | Yes             |
| Revenue lift observed in experiment        | \$308,259       |

**Decision: SHIP.** The result clears both bars — statistically significant ( $p=0.0024 < 0.05$ , confidence interval entirely positive) and business-significant (uplift exceeds the 1.0 percentage-point bar set for engineering/rollout cost). This is a rare 'ship with confidence' result — both statistical and business significance line up in the same direction.

## 6. RFM Customer Segmentation

Question: Who are our customers in terms of Recency, Frequency, and Monetary value, and what should we do for each group?



Total revenue contribution by RFM segment.

| Segment             | Customers | Total Value | % of Revenue | Avg Recency (days) |
|---------------------|-----------|-------------|--------------|--------------------|
| Champions           | 1,972     | \$293.0M    | 41.0%        | 22                 |
| Loyal Customers     | 2,549     | \$188.5M    | 26.4%        | 64                 |
| Cannot Lose Them    | 646       | \$74.0M     | 10.4%        | 252                |
| At Risk             | 1,094     | \$52.4M     | 7.3%         | 311                |
| Hibernating         | 1,025     | \$36.2M     | 5.1%         | 234                |
| Lost                | 1,913     | \$31.6M     | 4.4%         | 394                |
| New Customers       | 1,055     | \$28.4M     | 4.0%         | 25                 |
| Potential Loyalists | 182       | \$9.7M      | 1.4%         | 99                 |

Champions + Loyal Customers make up 43% of purchasing customers but drive 67% of total revenue from this customer base — protecting these two segments has an outsized ROI compared to broad-based acquisition spend.

### Recommended action per segment:

| Segment          | Recommended Action                                                     |
|------------------|------------------------------------------------------------------------|
| Champions        | VIP treatment, early access to new products, upsell premium items      |
| Loyal Customers  | Cross-sell complementary products, loyalty rewards program             |
| Cannot Lose Them | Urgent high-touch outreach — high past value, going cold               |
| At Risk          | Targeted win-back campaign with personalized offer before they're lost |
| Hibernating      | Low-cost reactivation email, otherwise low priority                    |

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| Lost                | Win-back campaign only if CAC-justified; otherwise deprioritize  |
| New Customers       | Onboarding sequence, first-purchase follow-up, education content |
| Potential Loyalists | Membership/loyalty program invite to increase frequency          |
| Others              | Monitor — doesn't fit a clean pattern yet                        |

## 7. Consolidated Recommendations

- Ship the checkout redesign — the A/B test result is statistically and business-significant with no caveats.
- Launch targeted retention outreach to the 407 customers on the high-value / high-churn-risk list before their 90-day churn window closes.
- Treat the 90-day linear revenue forecast as a floor, not a plan — build a growth-curve-aware forecast before using these numbers for budgeting, given how clearly the recent trend is accelerating past what a straight line predicts.
- Investigate the flagged anomaly dates (revenue, order volume, marketing spend, refunds) with the relevant team — these are candidates for real operational issues or one-off events worth understanding.
- Weight loyalty and VIP-style investment toward the Champions and Loyal Customers RFM segments, since they generate the large majority of revenue relative to their size — protecting them is cheaper than replacing them.
- Review the Cancelled/Pending order share with the checkout/fulfillment team — this is revenue that was attempted but never captured, separate from churn or marketing.

---

Methodology notes: all figures are derived from the processed dataset (customers, orders, order\_items, products, web\_events, marketing\_campaigns, returns, payments, experiments). Full detail tables (customer-level churn scores, RFM scores, forecast series, anomaly lists) are provided alongside this report as CSV files. Reference date used for recency-based calculations: 2025-12-31.